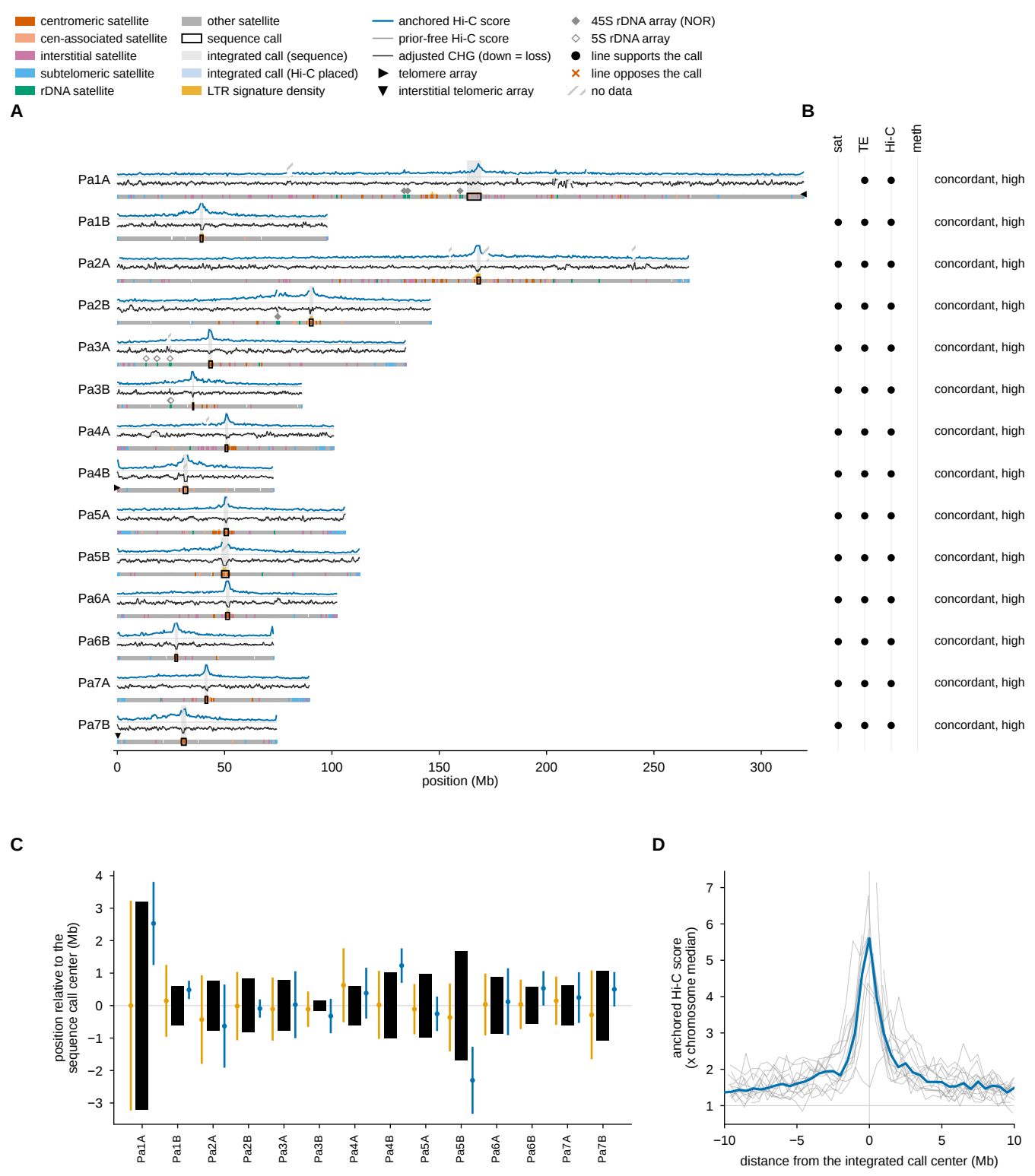
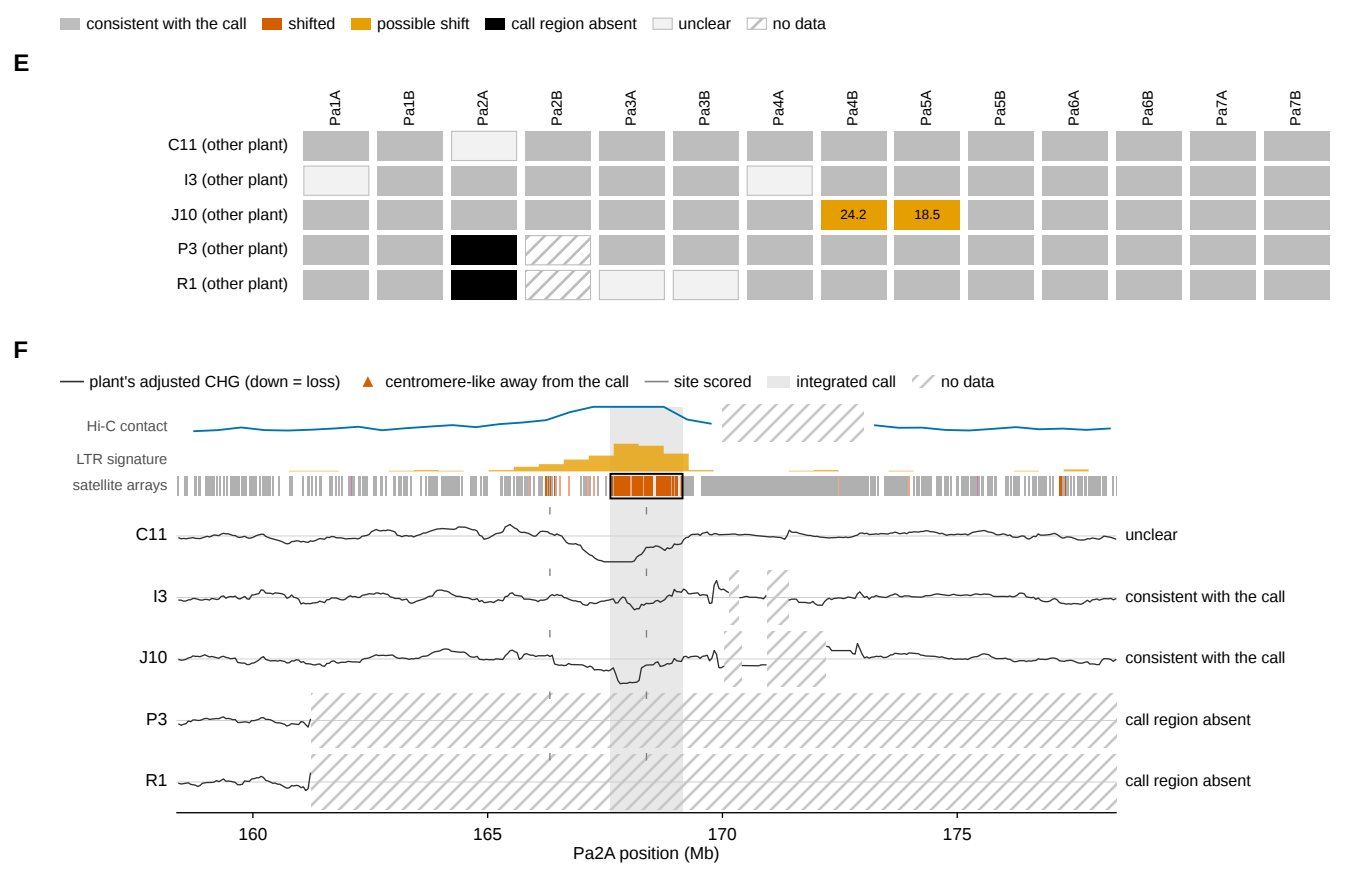

centromeric satellite
cen-associated satellite
interstitial satellite
subtelomeric satellite
rDNA satellite
other satellite
sequence call
integrated call (sequence)
integrated call (Hi-C placed)
LTR signature density
anchored Hi-C score
prior-free Hi-C score
adjusted CHG (down = loss)
telomere array
interstitial telomeric array
45S rDNA array (NOR)
5S rDNA array
line supports the call
line opposes the call
no data
A
B
sat
TE
Hi-C
meth
Pa1A
Pa1B
Pa2A
Pa2B
Pa3A
Pa3B
Pa4A
Pa4B
Pa5A
Pa5B
Pa6A
Pa6B
Pa7A
Pa7B
concordant, high
0
50
100
150
200
250
300
position (Mb)
C
position relative to the sequence call center (Mb)
4
3
2
1
0
−1
−2
−3
Pa1A
Pa1B
Pa2A
Pa2B
Pa3A
Pa3B
Pa4A
Pa4B
Pa5A
Pa5B
Pa6A
Pa6B
Pa7A
Pa7B
D
anchored Hi-C score (x chromosome median)
7
6
5
4
3
2
1
−10
−5
0
5
10
distance from the integrated call center (Mb)


consistent with the call
shifted
possible shift
call region absent
unclear
no data
E
Pa1A
Pa1B
Pa2A
Pa2B
Pa3A
Pa3B
Pa4A
Pa4B
Pa5A
Pa5B
Pa6A
Pa6B
Pa7A
Pa7B
C11 (other plant)
I3 (other plant)
J10 (other plant)
P3 (other plant)
R1 (other plant)
24.2
18.5
F
plant's adjusted CHG (down = loss)
centromere-like away from the call
site scored
integrated call
no data
Hi-C contact
LTR signature
satellite arrays
C11
I3
J10
P3
R1
unclear
consistent with the call
consistent with the call
call region absent
call region absent
160
165
170
175
Pa2A position (Mb)

In annua, 14 of 14 chromosomes have a centromere call; 0 sit away from (or without) a satellite array where Hi-C contact with the other centromeres and methylation place them, and 0 stay contested. Satellite position and centromere function can part ways, so each line is shown on its own.

A, Every chromosome at one scale: satellite arrays by family role, the density of the LTR signature's elements, Hi-C contact with the other centromeres and the methylation track, with the sequence call (box) and the integrated call (band).

B, Which evidence lines support or oppose each integrated call.

C, Where each line places the centromere relative to the satellite call.

D, Hi-C contact with the other chromosomes' centromeres peaks at the integrated calls (orange: calls placed away from the satellite).

E, Each of 5 plants, chromosome by chromosome: is its own evidence consistent with the reference call (gray), or does its centromere seem to sit elsewhere (orange; numbers: Mb from the call)? Black: the plant lacks the DNA of the call.

F, Pa2A, plant by plant: each plant's methylation track (centromeric loss points down) beneath the reference plant's evidence; triangles mark where a plant looks centromeric away from the call.

Terms. Sequence call: the satellite (or retrotransposon) centromere. Anchored Hi-C score: mean observed/expected contact of each bin with windows at the other chromosomes' sequence calls (the chromosome's own call is never used), over the chromosome median. Integrated call: the sequence call, unless Hi-C that agrees with the sequence calls of the other chromosomes (gate: majority, Poisson-binomial P <= 0.01) places a strong peak elsewhere; a high sequence call moves only with a second line (retrotransposon core or methylation) at the Hi-C peak. Where that gate fails, the arm fold (the anti-diagonal of each chromosome's own contact map in a Rabl nucleus) can validate the prior-free Hi-C peaks instead (majority, P <= 0.005), or the trans arm alignment (loci at equal distances from their centromeres meet across chromosomes; P <= 0.005, where the fold leans the same way, P <= 0.05); the two 3D lines then move calls they both contradict. Lines used here: satellite, te, hic, meth.

A, 14 chromosomes in assembly order, to one base-pair scale. Each satellite array is a box colored by its family's role in this genome (36,770 arrays; arrays too small to see are drawn at the minimum visible width). The orange area is the density of the CRM LTR signature (the marker's families). Blue: anchored Hi-C score at 500-kb bins (0 to 4.5 x the chromosome median over the track height, clipped above); gray: the prior-free Hi-C score. The gate passed on 14 of 14 chromosomes; the anchored peak lies on the sequence call on 14 of 14, the prior-free peak on 14. Black line: composition-adjusted CHG methylation pooled over 5 plants (0 = like the same sequence class nearby; down = loss). The centromere signature learned from the calls of the other chromosomes: CHG LTR:CRM down,CG LTR:CRM down. Triangles: telomere arrays of the unit AAACCCT (found de novo; 2 of 28 chromosome ends); downward triangles: 1 array of the unit inside chromosomes (interstitial telomeric sequences). Gray diamonds: rDNA arrays, 4 of 45S (filled; the 18S motif in tandem) and 6 of 5S (open; whole 5S genes in tandem). Hatching: no data (masked or uncovered bins). Bands: integrated calls; arrows: sequence call to moved call.

B, One column per evidence line: filled circle, the line supports the integrated call (satellite: >= 5 kb of centromeric-family arrays inside it; TE: the marker core agrees; Hi-C: gated (or, on the support-only test, significant) and the anchored peak within one bin, or the prior-free peak that placed the call; methylation: centromere-like under the leave-one-chromosome-out signature; fold and align, shown only where they acted: the call lies in the arm-fold or the trans arm-alignment core); cross, a gated strong Hi-C peak, a decisive LTR core, the arm fold or alignment, or a validated prior-free Hi-C peak points elsewhere. Text: relation and confidence.

C, Black bar: the sequence call, centered at 0. Orange: the TE marker core; blue: the anchored Hi-C core with its peak; open diamond: the integrated call when it differs from the sequence call.

D, Anchored Hi-C score (prior-free score where no sequence call anchors it) within 40 Hi-C bins (at most 10 Mb) on each side of every integrated call, one trace per chromosome (gray: kept sequence calls; orange: calls placed by Hi-C); blue: the median. A peak at 0 means the calls sit where the chromosomes' centromeres meet in the nucleus.

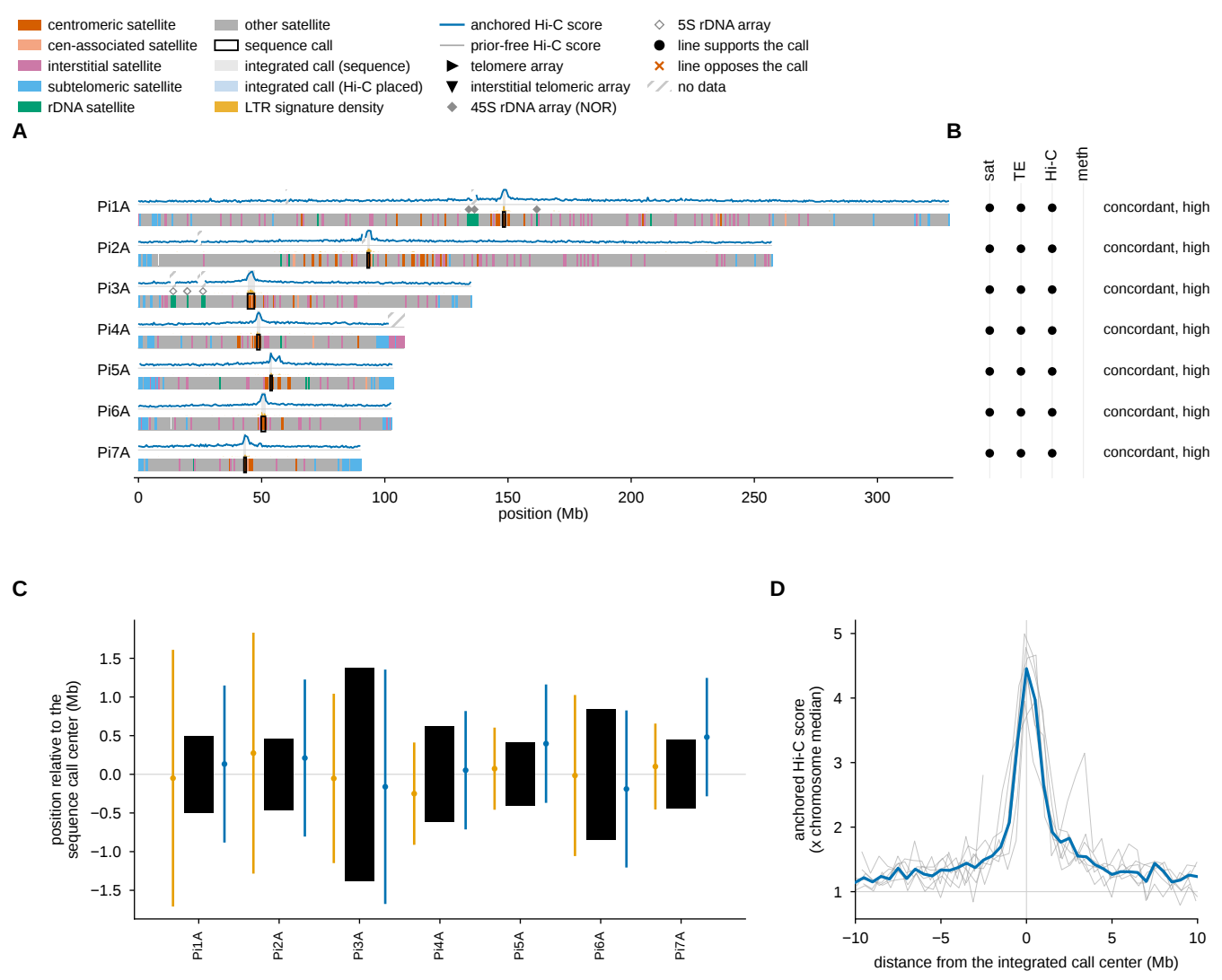
Plants and provenance. Each input was tested for whether it comes from the reference plant (the assembled genotype): methylation samples by fixed differences (symmetric CHG sites whose rows are missing on both strands where reads span them, so that read errors, which strike one strand, do not count) that another sample lacks; every input by segments it lacks while another input covers them. 5 of 5 plants come from another plant (C11, I3, J10, P3, R1); only reference-plant inputs and inputs of unknown origin decide the reference calls (methylation samples deciding here: .).

E, One row per plant (its provenance in brackets), one column per chromosome. Gray: consistent with the reference call (centromere-like methylation there, or the plant's own Hi-C peak there or where the reference library peaks); methylation that looks centromeric is not proof, since some non-centromeric places look so in every plant. Orange: a dip at the call and centromere-like methylation at a site away from it, with between-plant support (the plant less centromeric than the others at the call, or more at that site; signature-score differences against their spread at the other chromosomes' calls, Student's t prediction interval, Benjamini-Hochberg q <= 0.05 over 123 tests), or a strong Hi-C peak of the plant's own library away from the call where the reference library peaks elsewhere. Light orange: the methylation pattern without between-plant support. Black: the call lies in a segment the plant lacks. Purple: the plant's lines disagree. Pale: unclear; hatched: no data at the call. Of 70 cells: 59 at the call, 0 shifted, 2 possible shifts, 2 call regions absent, 0 with disagreeing lines, 5 unclear, 2 without data, 0 on chromosomes without a reference call.

F, Pa2A, chosen as the chromosome with the most shift verdicts. Top: the reference plant's evidence (satellite arrays by family role, the retrotransposon marker density in orange, the anchored Hi-C score in blue, the sequence call boxed); the band is the integrated call. Below, one row per plant: its composition-adjusted CHG track (one scale for all plants, +-0.16; down = loss), gray ticks at the sites every plant was scored at, a triangle at its best centromere-like site away from the call, and its verdict at the right. Hatching: no data (masked bins or DNA the plant lacks).

Exceptions are listed in annua.centromeres.tsv (relation, against, flags, seq_flags) and every candidate feature in annua.loci.tsv.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11/v12/v13/v14/runs/rel15b/ALL/poa · Script: cenfuse/plot.py · 2026-11-08


centromeric satellite
cen-associated satellite
interstitial satellite
subtelomeric satellite
rDNA satellite
other satellite
sequence call
integrated call (sequence)
integrated call (Hi-C placed)
LTR signature density
anchored Hi-C score
prior-free Hi-C score
telomere array
interstitial telomeric array
45S rDNA array (NOR)
5S rDNA array
line supports the call
line opposes the call
no data
A
Pi1A
Pi2A
Pi3A
Pi4A
Pi5A
Pi6A
Pi7A
0
50
100
150
200
250
300
position (Mb)
B
sat
TE
Hi-C
meth
concordant, high
C
position relative to the sequence call center (Mb)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Pi1A
Pi2A
Pi3A
Pi4A
Pi5A
Pi6A
Pi7A
D
anchored Hi-C score (x chromosome median)
5
4
3
2
1
−10
−5
0
5
10
distance from the integrated call center (Mb)

In infirma, 7 of 7 chromosomes have a centromere call; 0 sit away from (or without) a satellite array where Hi-C contact with the other centromeres place them, and 0 stay contested. Satellite position and centromere function can part ways, so each line is shown on its own.

A, Every chromosome at one scale: satellite arrays by family role, the density of the LTR signature's elements, Hi-C contact with the other centromeres and the methylation track, with the sequence call (box) and the integrated call (band).

B, Which evidence lines support or oppose each integrated call.

C, Where each line places the centromere relative to the satellite call.

D, Hi-C contact with the other chromosomes' centromeres peaks at the integrated calls (orange: calls placed away from the satellite).

Terms. Sequence call: the satellite (or retrotransposon) centromere. Anchored Hi-C score: mean observed/expected contact of each bin with windows at the other chromosomes' sequence calls (the chromosome's own call is never used), over the chromosome median. Integrated call: the sequence call, unless Hi-C that agrees with the sequence calls of the other chromosomes (gate: majority, Poisson-binomial P <= 0.01) places a strong peak elsewhere; a high sequence call moves only with a second line (retrotransposon core or methylation) at the Hi-C peak. Where that gate fails, the arm fold (the anti-diagonal of each chromosome's own contact map in a Rabl nucleus) can validate the prior-free Hi-C peaks instead (majority, P <= 0.005), or the trans arm alignment (loci at equal distances from their centromeres meet across chromosomes; P <= 0.005, where the fold leans the same way, P <= 0.05); the two 3D lines then move calls they both contradict. Lines used here: satellite, te, hic.

A, 7 chromosomes in assembly order, to one base-pair scale. Each satellite array is a box colored by its family's role in this genome (29,423 arrays; arrays too small to see are drawn at the minimum visible width). The orange area is the density of the CRM LTR signature (the marker's families). Blue: anchored Hi-C score at 500-kb bins (0 to 3.8 x the chromosome median over the track height, clipped above); gray: the prior-free Hi-C score. The gate passed on 7 of 7 chromosomes; the anchored peak lies on the sequence call on 7 of 7, the prior-free peak on 7. No telomere unit was enriched in tandem arrays at the chromosome ends. Gray diamonds: rDNA arrays, 3 of 45S (filled; the 18S motif in tandem) and 4 of 5S (open; whole 5S genes in tandem). Hatching: no data (masked or uncovered bins). Bands: integrated calls; arrows: sequence call to moved call.

B, One column per evidence line: filled circle, the line supports the integrated call (satellite: >= 5 kb of centromeric-family arrays inside it; TE: the marker core agrees; Hi-C: gated (or, on the support-only test, significant) and the anchored peak within one bin, or the prior-free peak that placed the call; methylation: centromere-like under the leave-one-chromosome-out signature; fold and align, shown only where they acted: the call lies in the arm-fold or the trans arm-alignment core); cross, a gated strong Hi-C peak, a decisive LTR core, the arm fold or alignment, or a validated prior-free Hi-C peak points elsewhere. Text: relation and confidence.

C, Black bar: the sequence call, centered at 0. Orange: the TE marker core; blue: the anchored Hi-C core with its peak; open diamond: the integrated call when it differs from the sequence call.

D, Anchored Hi-C score (prior-free score where no sequence call anchors it) within 40 Hi-C bins (at most 10 Mb) on each side of every integrated call, one trace per chromosome (gray: kept sequence calls; orange: calls placed by Hi-C); blue: the median. A peak at 0 means the calls sit where the chromosomes' centromeres meet in the nucleus.

Exceptions are listed in infirma.centromeres.tsv (relation, against, flags, seq_flags) and every candidate feature in infirma.loci.tsv.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11/v12/v13/v14/runs/rel15b/ALL/poa · Script: cenfuse/plot.py · 2026-11-08